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Gradient descent

The loss is minimised by stepping against the gradient. For a learning rate $\eta > 0$ the update is

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta L(\theta_t)$$

and for mean squared error $L(\theta) = \frac{1}{n}\sum_{i=1}^{n} (y_i - \theta^\top x_i)^2$ the gradient is